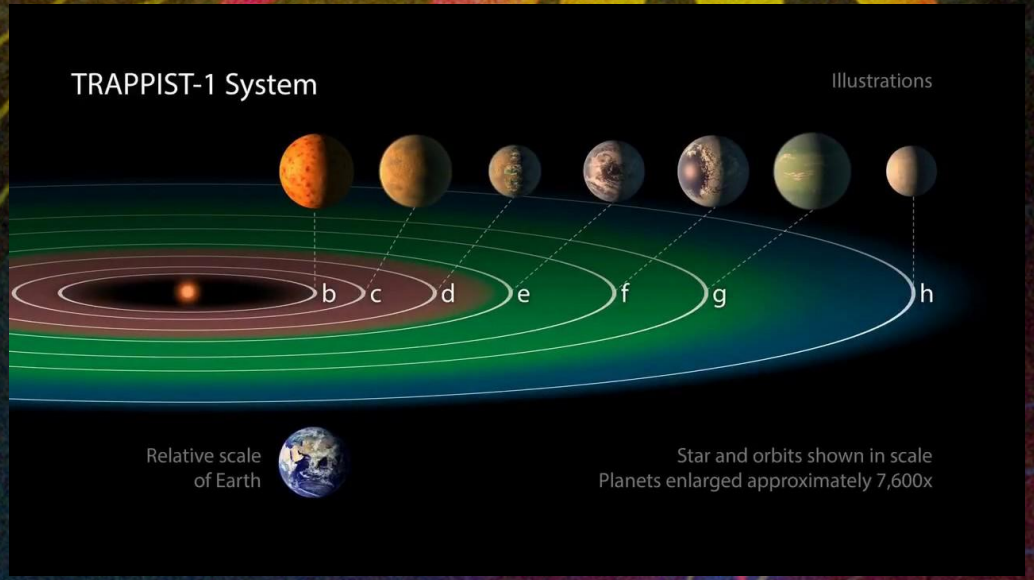
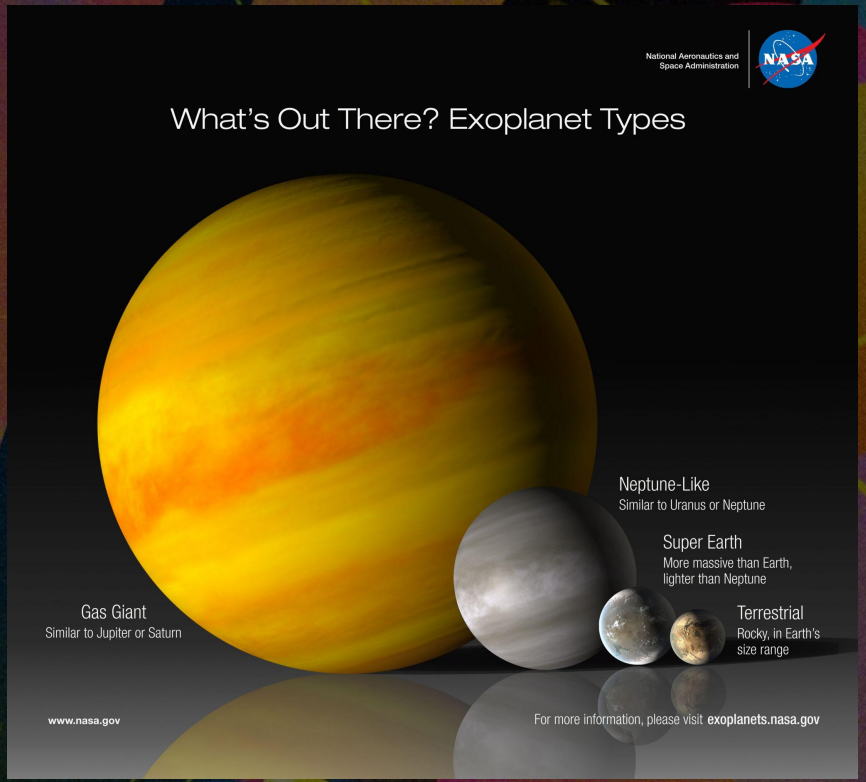
Exoplanets

Aubrie Thoms



What are Exoplanets?



- "An exoplanet is any planet beyond our solar system. Most orbit other stars, but free-floating exoplanets, called rogue planets, orbit the galactic center and are untethered to any star." (NASA)
- Types: Gas giant, Neptunian, Super-Earth, Terrestrial

In Today's Lesson...

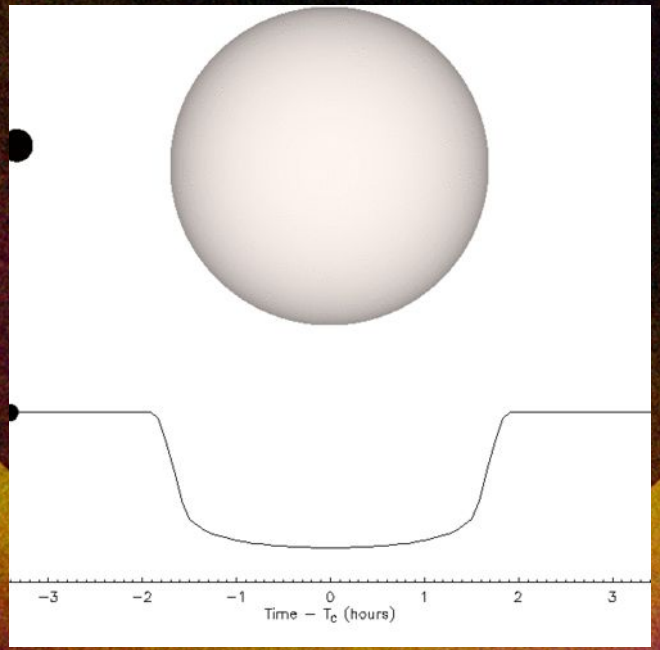
You will:

- ★ Be introduced to the transit method
- ★ Be able to apply your knowledge of exoplanets to a new concept
- ★ Be able to identify the basic components of a transit curve

An Exoplanet Transits When...

It passes between us (the observer) and its host star!

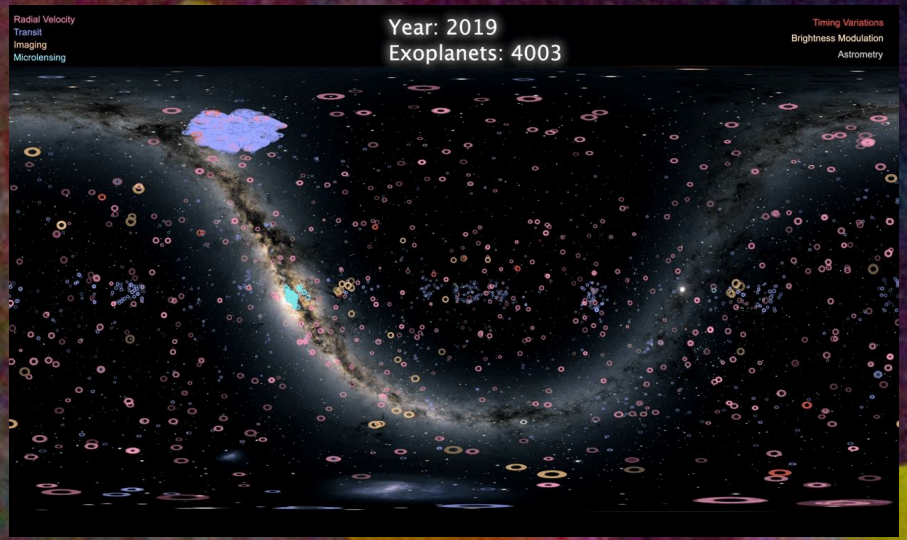
- ★ Most exoplanets have been discovered through the transit method
- ★ Helps with characterizing exoplanets
- ★ In the GIF: Planet is transiting, blocking the light from star. The graph displays the change of brightness over time.



How can we benefit from the transit method?

- ★ Size
  - “For example, if the starlight dims by 1 percent, this indicates that the planet is 1 percent the size of its star.” (MIT)
  - Radius, Mass, Period/Orbit, etc.

Q: What year was the first discovery of an exoplanet through the transit method?



a) 1956
b) 2002
c) 1978
d) 2015

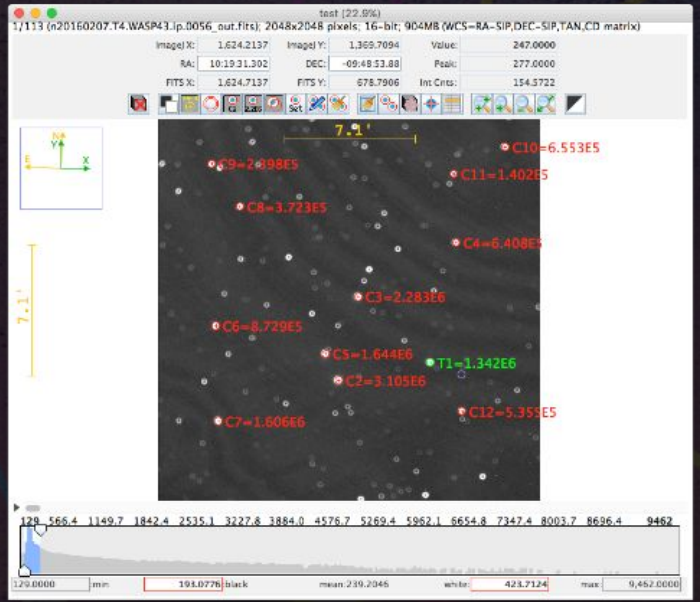
A Brief History

- ★ 2002 - OGLE-TR-56b was the first exoplanet to be discovered through the transit method
- ★ 1999 - HD 209458 b was the first exoplanet to be found with the transit method
  - ○ ...Already known from radial velocity
- ★ Radial velocity was the primary method for detection until the turn of the century



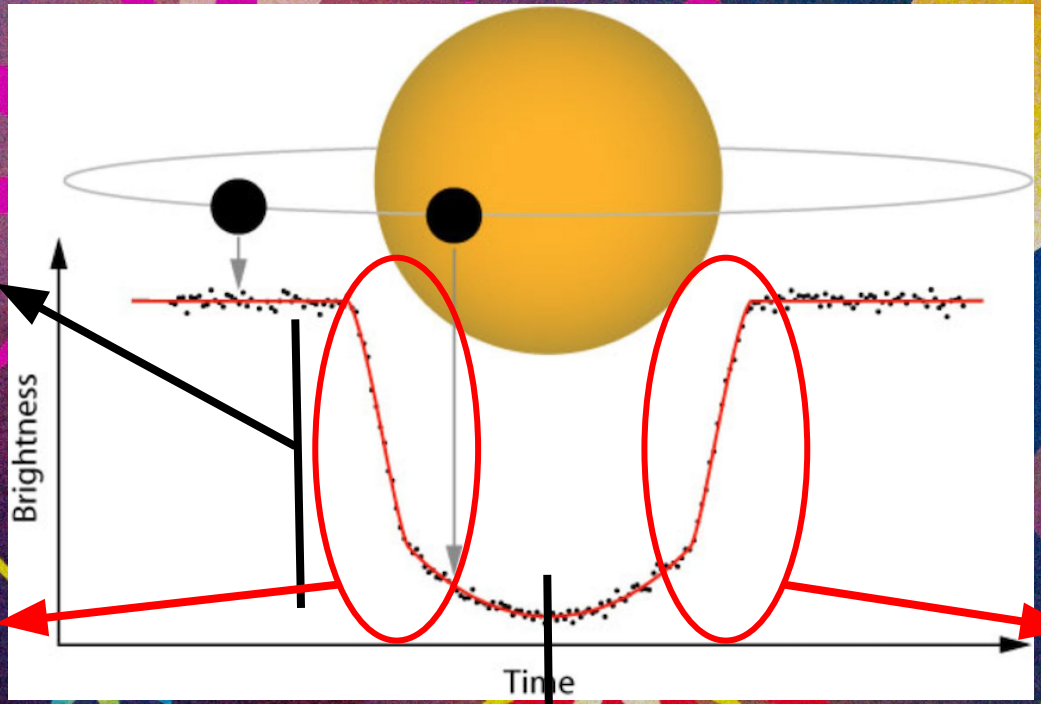
OGLE-TR-56b size comparison to Jupiter:
Has a radius of 1.36 R_{Jup} and a mass of 1.38 M_{Jup}

Transit Method SparkNotes



1. Observations
   a. Ground-based telescopes: CCD, or camera
2. Data calibration
   a. Clean up those images!
3. Photometry
   a. Measures the brightness of a star in an image
   b. Comparison stars -> used as a reference for brightness
   c. Transit curve -> Brightness over time

Transit Curve



Transit depth: Shows the amount of light "blocked" from the star from the planet

Ingress: Entering contact, blocking light from the star's disk/edge

Egress: Exiting contact, blocking light from the star's disk/edge (opposite of ingress)

Transit center: Peak of the transit, maximum limit of coverage/dimmest

Try It Yourself!

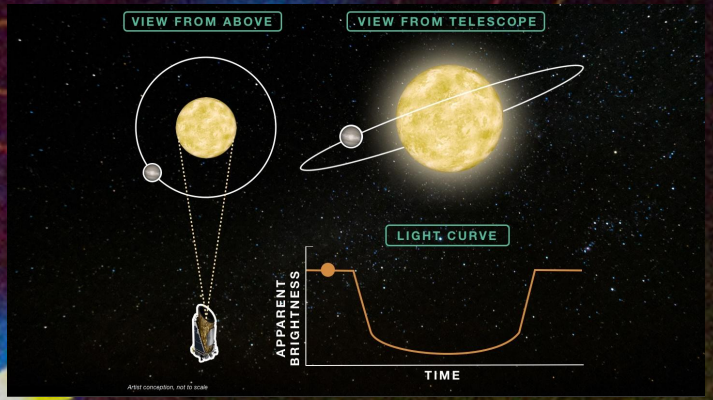


Answers:
1b,2c,3a

Some hints:

- ★ **Size of Star > Planet:**
  - **Shallow transit curve**
- ★ **Size of Planet > Star:**
  - **Deeper transit curve**
- ★ **Transit length is dependent on the orbit as well**
  - **Farther from the host star (semi-major axis) = larger orbit = larger period = longer transit time**
  - **Pay attention to scaling!**

To Conclude:





Now you can understand and identify transits!

Exoplanets are...

Planets NOT in our solar system

Transits are...

When the planet passes between its host star and the observer

We find transits by...

Observing a star and graphing its brightness over time

Transits can differ based on...

Planet's size, its star's size, and distance from each other


Questions?

Links

- What's a transit? – Exoplanet Exploration: Planets Beyond our Solar System (nasa.gov)
- 5 Ways to Find a Planet | Explore – Exoplanet Exploration: Planets Beyond our Solar System (nasa.gov)
- 7.4: The Transit Method - K12 LibreTexts
- Rains of Terror on Exoplanet HD 189733b - NASA
- Exoplanet: TRAPPIST - 1b - Science On a Sphere (noaa.gov)
- Hot Jupiter WASP-12b is Spiraling In Toward Its Star: Study | Sci.News
- Light curves of the long-slit analysis of HD 189733b and the... | Download Scientific Diagram (researchgate.net)
- Folded light curve of WASP-12b. | Download Scientific Diagram (researchgate.net)
- Transit of TRAPPIST-1b observed with the ARCTIC instrument (black and... | Download Scientific Diagram (researchgate.net)
- WASP-12 | NASA Exoplanet Archive (caltech.edu)
- TRAPPIST-1 | NASA Exoplanet Archive (caltech.edu)
- HD 189733 | NASA Exoplanet Archive (caltech.edu)
- ESA - Exoplanets in the spotlight
- Overview | Planet Types – Exoplanet Exploration: Planets Beyond our Solar System (nasa.gov)
- Exoplanet-catalog – Exoplanet Exploration: Planets Beyond our Solar System OGLE-TR-56 b (nasa.gov)
- Overview | What is an Exoplanet? – Exoplanet Exploration: Planets Beyond our Solar System (nasa.gov)
- How We Find and Characterize | Discovery – Exoplanet Exploration: Planets Beyond our Solar System (nasa.gov)
- Transit Method - Las Cumbres Observatory (lco.global)
- Explained: Transiting exoplanets | MIT News | Massachusetts Institute of Technology
- Introduction to Photometry - Las Cumbres Observatory (lco.global)
- ESA Science & Technology - Exoplanet detection methods